Software Requirements Specification
Software Engineering

Team #14, Troitsky Toolbox
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Table 1: Revision History

Date	**Developer(s)**	**Change**
10/02/2026	Pedro	SRS template (Meyer) adopted; section files created.
10/07/2026	Pedro	G.1 drafted from problem statement.
10/08/2026	Jack	P.6 risks and mitigation analysis added.
10/11/2026	Jack	P.6 expanded.
10/13/2026	Pedro	Formatting aligned with other team documents.

This document follows the requirements documentation structure presented in the Handbook of requirements and business analysis, by Bertrand Meyer.

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G Goals

Comment: *Goals are "needs of the target organization, which the system will address". While the development team is the principal user of the other books, the Goals book addresses a wider audience: essentially, all stakeholders.*

G.1 Context and overall objective

Comment: *High-level view of the project: organizational context and reason for building a system.*

The McMaster Troitsky club is a student-run organization that registers and supports 4-6 teams of students as they take on the task of modelling, analyzing and manufacturing a bridge out of popsicle sticks, white glue and dental floss. While the club has been successful in the past, results often vary in performance both between years and teams, due to the lack of centralized information and technical support for members. As teams progress through several years of competition, undergoing material testing, bridge design and analysis, and even experimenting with more innovative concepts, skill and knowledge within the club often consolidates within more experienced members or single teams, ensuring that when those members graduate, information is lost to the club. This project seeks to create a centralized information repository and technical support system for the McMaster Troitsky club, ensuring information transfer between years and teams, and facilitating support and onboarding for new members.

G.2 Current situation

Comment: *Current state of processes to be addressed by the project and the resulting system.*

Requirement 1. *This is a requirement example. It refines* **??**

G.3 Expected benefits

Comment: *New processes, or improvement to existing processes, made possible by the project's results.* **Comment:** *This section should not be empty!*

G.4 Functionality Overview

High Level functional requirements for the toolbox are defined below:

- **Data Storage and Retrival.** Troitsky Toolbox must store past designs, materials testing informat, and relevant development information. Troitsky toolbox must be able to return relevant data quickly based on user searches. The system should allow for the addition of new information, streamilining the process of growing the repository.

- **Key Model Analysis Features and Bridge Predictions.** Provide a platform for teams to upload their bridge models through a standardized format, allowing the user to view and analyze the model, highlighting key metrics and predictions about the bridge's performance. This should decrease the barrier to entry to bridge analysis, allowing all members of a team to get involved in the analysis process, even without formal modelling knowledge. This will be integrated with the material testing support to provide more accurate bridge analysis from the stored material properties.
- **Materials Testing Support and Facilitated Result Analysis** As the McMaster Troitsky club begins to standardize knowledge collection, material testing data will be collected and stored in a standardized format. The service will streamline this process, allowing testing results to be easily uploaded and analyzed against past results, along with collecting supplementary testing information for future years.
- **Simplified and Comprehensive Knowledge Transfer** Support documentation from the McMaster Troitsky club through exit documents and other knowledge transfer methods, ensuring that information and designs are consolidated and accessible to future members. Specialized member roles, such as executive positions, will also be able to keep a record of their role responsibilities and key factors, to further improve the club organization.

G.5 High-level usage scenarios

Comment: *Fundamental usage paths through the system.*

G.6 Limitations and exclusions

Below is a list of limitations for the system, items that are currently out of scope for development:

- The system is not looking to replace the more complex modelling capabilities of external/industry standard modelling software. Unlike the current apps being used by teams, such as AutoCAD Inventor, SAP2000 and SolidWorks, the modelling component within Troitsky Toolbox will be intentionally simplified, which will leave many of the more advanced features out of scope.
- The system will not support user access for past members. The service will only be accessible by current members, with only the public-facing landing page and gallery available to others.
- The system will not replace all forms of communication between members. Given the accepted use of other platforms for communication and the ease of use in utilizing an existing app for communication, peer-to-peer or individual team channels will not be supported in the toolbox.

- The system will not replace registration into the club; initial applications into specific roles will remain external. Payment support will be excluded and done externally from the service with registration.

G.7 Stakeholders and requirements sources

Comment: *Groups of people who can affect the project or be affected by it, and other places to consider for information about the project and system.* Comment: *This section should not be empty!*

E Environment

Comment: *The Environment book describes the application domain and external context, physical or virtual (or a mix), in which the system will operate.*

E.1 Glossary

Comment: *Clear and precise definitions of all the vocabulary specific to the application domain, including technical terms, words from ordinary language used in a special meaning, and acronyms. This chapter should not be empty!*

E.2 Components

The system will interact with the following external components:

- **Official Troitsky Organization - Concordia:** As the Troitsky club is generally officiated and organized fully by Concordia University's Troitsky organization, much of what is valued in the product depends on what is valued in the competition itself. Between years, rules are often revised, and different competition components are added, removed or modified. As is the case, the product will be heavily dependent on the organization and must allow for compliance and rule checking in line with the current competition standards.
- **CAD Software Modelling Tools:** This is used by teams to create their STEP files, which will serve as the input for the Model Analysis.
- **Static Model Analysis APIs:** For the modelling component of the product, external APIs will be used to perform the static analysis of the inputted STEP files. This will be used to calculate/create the output predictions and model data.
- **Hosting and Data Storage Platform:** The product will require a platform or server to host the website and the shared database. This is where the repository of information, uploaded files and backups will be stored.

- **Materials Testing Department Setup:** As the product seeks to support the material testing department, the app will need to work around the data formats and the processing the team prefers.

- **Club Sponsors and University Funding:** The MES and external sponsors dictate the funding that the club receives yearly. This funding may dictate the future upkeep of the product, being required for the hosting and data storage platform and potentially the external modelling API usage.

E.3 Constraints

Troitsky Toolbox Constraints

- **Competition Rules:** The model compliance checker and other support services within Troitsky Toolbox must comply with the rules set out for the competition. Outdated guides must be labelled to ensure users understand they do not fit with current competition rules.

- **Member Data and Privacy:** As the service seeks to compile team and member information, member privacy and sharing permissions dictate what can be collected. During the current competition year, individual team files must be left separate, with only executives and coaches being able to view all files. After the competition is complete, exit documentation collected will determine what individuals wish to have stored for future years.

- **Club Data Format Compatibility :** The system must accommodate the file formats that the club utilizes, being constrained to a generalized file format when multiple are available. In the case of modelling files, STEP (.stp) files must be used as input as they are available across utilized services. The material testing data collection must accept .csv files, working with the data exporting the club utilizes.

Operational Constraints

- **Budget:** The funding set out by the MES, along with the yearly club sponsorships, will determine the budget and therefore constrain the services and upkeep of the product. Troitsky Toolbox must operate within its share of the overall club funds, dictating the storage capabilities and API usage of the service.

- **Troitsky Software Subteam:** Maintenance year to year after this year will be limited by the number of software subteam members the club recruits. As the subteam is made up of student volunteers, the experience and drive of the members will also determine the level of development of the system.

E.4 Assumptions

The following properties of the environment are assumed.

- **Competition Format Stability:** Despite the product being adaptable to changes within rules and smaller operational changes, the overall format of the Troitsky competition is assumed to remain stable across years. While some forms of bridges, materials or scoring are revised, the product serves the standard competition format of popsicle stick, white glue and floss bridges tested for ultimate point load at the centre. The rule set may change across years, but competition rulebooks are assumed to be published at the start of each competition year, allowing for integration into the rule compliance checker once yearly, with only minor modifications necessary.

- **User Access and Device Availability:** As the product will be available through a website, there is an assumption that members will both have access to a suitable device (desktop or laptop) and a standard internet browser. There is also an assumption that all users will have a working internet connection when accessing the service.

- **McMaster Troitsky Club Organizational Structure:** The role-based access control of the system will be designed around the current club structure, specifically around the roles in the organizational body of the club (general members, team captains, coaches, executives). It is assumed that this structure will not change frequently and that the access of each role will remain constant across competition years.

- **Continued Club Operation:** There is an assumption that the Troitsky competition, both as a whole and as McMaster's individual club, will continue to operate. This includes the assumption that teams will continue to register and that the software subteam will continue to have registered members.

- **Model File Conventions:** Bridge models are assumed to be exported in the STEP file format. This export will be in line with a standard export format outlined by the club, allowing for some consistency in how bridge members and units are represented across files. In line with P.6, these file conventions are linked to the most complex and high-risk component of the product, so an agreed-upon standard simplifies the scope of the service, serving as a mitigation strategy for the risk.

- **Material Behaviour:** There is an assumption of some consistency in material properties while the club purchases materials from the same supplier. As the model analysis predictions may rely on the measured material properties of popsicle stick and glue slabs, constant material properties are a required assumption to create any form of performance prediction.

E.5 Effects

Comment: *Elements and properties of the environment that the system will affect.*

- Centralization of development information - the wealth of development information (e.g. past designs, material testing information) will be in one central location for future members to access, which will facilitate onboarding.
- Model analysis - team members will be able to submit their models and recieve key metrics and predictions regarding that model through an automatic analysis.
- Facilitated comparison - team members will be able to easily compare modelling results for different models. This will facilitate development because team members will be able to start from the best and abandon changes that caused the model to degrade.
- Simplified knowledge transfer - veteran team members will be able to share their vast knowledge with new members more easily. This is critical, especially for outgoing team members.
- Creation of a permanant software sub-team - this project will be the genesis of a permanant sofware subteam of the club. Furthermore, brief but top-tier resources alongside the software suitewill be developed to smooth the onboarding of new software subteam members.

E.6 Invariants

Comment: *Properties of the environment that the system's operation must preserve.*

- The knowledge entered into the system, including development knowledge, scientific and engineering methods, and veteran wisdom, shall be fully preserved.
- Any model entered into the system, whether developed before or after the system's released, shall be fully preserved and not corrected at all to preserve development history as faithfully as possible for easy comparison.
- The rules of the competition that are relevant to the system shall be fully preserved and not be mutated at all. This is because modelling will be based of the rules as they are, and any deviation from the rules will result in inferior models.

S System

Comment: *The System book refines the Goal one by focusing on more detailed requirements about the system under development, mainly its constituents, behaviors and properties.*

S.1 Components

Comment: *Overall structure expressed by the list of major software and, if applicable, hardware parts.* **Comment:** *This section should not be empty!*

S.2 Functionality

Comment: *One section, S.2.n, for each of the components identified in S.2, describing the corresponding behaviors (functional and non-functional properties).* **Comment:** *This section should not be empty!*

S.3 Interfaces

Comment: *How the system makes the functionality of S.2 available to the rest of the world, particularly user interfaces and program interfaces (APIs).*

S.4 Detailed usage scenarios

Comment: *Examples of interaction between the environment (or human users) and the system: use cases, user stories.*

S.5 Prioritization

Comment: *Classification of the behaviors, interfaces and scenarios (S.2, S.3 and S.4) by their degree of criticality.*

S.6 Verification and acceptance criteria

Comment: *Specification of the conditions under which an implementation will be deemed satisfactory.*

P Project

Comment: *The Project book describes all the constraints and expectations not about the system itself, but about how to develop and produce it.*

P.1 Roles and personnel

P.1.1 Project Supervisor - Mason

The project supervisor will be responsible for coordinating internal deadlines, meetings and development timelines. The supervisor will serve as the main point of contact with administration, the supervisor and the TA, and serve as the default team lead.

P.1.2 Troitsky Club Communication Associate - Pedro

The Troitsky Club Communication Associate will be responsible for the majority of the coordination with the club organization and executive members. Surveys and interviews will be organized, distributed, and run by this member.

P.1.3 Front-end / UI Developer - Pedro, Michael

This role will be responsible for the UI of the product, both in technical displays of data and in more theme/design heavy components. They will build the web app's interface, so experience in web frameworks would be beneficial. They will be the ones primarily responsible for the usability testing in the Winter term's CD.

P.1.4 Back-end/Data Developer - Jack, Mason

This role will be responsible for the design of the data storage within the app. This will cover both the files and data the club stores (bridge files, tutorials), but also RBAC information and user/club site preferences.

P.1.5 Quality Assurance / Tester - Michael

The Quality Assurance / Tester will be responsible for maintaining a standard of quality in both code and documentation of the web app. This will involve a review of final product content and the overall design of the testing within the system.

P.1.6 Supervisor - Isabella Vescio

The supervisor for the project will provide guidance, domain expertise and general feedback for the service. As they are heavily involved in the club, they will also support interaction and integration with the club.

P.1.7 Teaching Assistant - Mohamad Mirhoseininejad

The TA will provide course-related feedback, answer administrative questions and provide feedback on expectations and course grading. They will serve as the main point of contact with course administration. General project feedback during TA meetings will also help keep the project timeline in check.

P.2 Imposed technical choices

Comment: *Any a priori choices binding the project to specific tools, hardware, languages or other technical parameters.*

P.3 Schedule and milestones

Comment: *List of tasks to be carried out and their scheduling.* Comment: *This section should not be empty!*

P.4 Tasks and deliverables

Comment: *Details of individual tasks listed under P.3 and their expected outcomes.* Comment: *This section should not be empty!*

P.5 Required technology elements

Comment: *External systems, hardware and software, expected to be necessary for building the system.*

P.6 Risks and mitigation analysis

Comment: *Potential obstacles to meeting the schedule of P.4, and measures for adapting the plan if they do arise.*

- **Risk: 3D Model Import and Analysis is Infeasible**
  - **Description:** Parsing STEP (.stp) files, rendering them, and extracting member geometry for the rule compliance and performance prediction tools is the most technically complex feature of the Toolbox. Available libraries may not support the required data extraction, or may not run in a web environment.
  - **Mitigation:** Prove the import and visualization pipeline in the Proof of Concept demonstration before dependent tools are built. Use an established geometry library rather than a custom parser, and agree on a standard CAD export template with the club to limit the variety of input files.
  - **Contingency:** Fall back to manual or spreadsheet entry of member geometry so the rule compliance and prediction tools remain usable without 3D import.
- **Risk: Insufficient Legacy Club Data**
  - **Description:** Past designs, material testing results and documentation are scattered across members and inconsistent in format, so the information repository may launch with little useful content.
  - **Mitigation:** Collect existing documents with the club executives early in the project and define standard upload templates for designs and material tests.
  - **Contingency:** Launch with the current season's data only, and plan for the repository to be populated over time through the upload tools.

- **Risk: Technical Issues with External Libraries**
  - **Description:** Technical issues may arise with external technologies used by tools such as bridge modelling and rule compliance, including licensing or deployment limits (for example, Autodesk APIs requiring a desktop installation).
  - **Mitigation:** Confirm licensing and deployment constraints of each library before adopting it. Team members test and validate their code before opening a pull request, and at least one other member reviews it.
  - **Contingency:** Replace the library with an open-source alternative, or reduce the affected tool's scope.
- **Risk: Unrealistic Schedule or Workload**
  - **Description:** Tasks may take longer than expected or workload may become unbalanced among team members.
  - **Mitigation:** Track tasks and milestones on the GitHub Kanban board. The Notetaker records timelines at check-in meetings, and the team revises estimates and reassigns tasks as needed.
  - **Contingency:** Because tools are largely independent, defer lower-priority tools to the software subteam's backlog rather than delaying core features.
- **Risk: User Interface is Not User-Friendly**
  - **Description:** The interface may not be intuitive for Troitsky members with varying levels of technical and modelling experience.
  - **Mitigation:** Run usability testing sessions with both new and experienced members (see the Usability Testing CD), gather feedback, and make iterative improvements.
  - **Contingency:** Prioritize fixes to the most-used workflows and add in-app guidance for complex tools.

P.7 Requirements process and report

McMaster Troitsky Executives
Both Pedro Azevedo and our supervisor Bella Vescio are Troitsky Executives; our requirements will be gathered through team meetings and supervisor meetings.
Important Questions to Ask:

- What information is most important to consolidate?
- When taking your position, what would you have liked to have seen within a transition document?

- How would you go about outlining the duties of a software development and support team?

The consolidation of information and ease of transition is one of the main goals in the development of this project, so we would want to hear the executives' opinions on this. Additionally, the challenge of this project is the maintainability and expandability of the code, so we must have an understanding of how future software teams are set up so that we can prepare documentation accordingly.

General McMaster Troitsky Members
A survey will be sent to general Troitsky members to elicit the overall wants and needs of members.
Questions in the survey will be:

- How many years of experience do you have in Troitsky (1-5 years)
- What is your current role?
- What is your modelling experience?
- How valuable would you find an entry-level modelling service, like implementing simple testing and weak point analysis?
- Rate (1-5) how valuable you would find having this information easily accessible regarding past McMaster Troitsky data. [**Bridge Modelling files, Joint Construction Data (specific model use), Manufacturing information, Past Team Results, Past Technical Presentation Material**]
- How does your team store data [**Microsoft Teams, Google Drive, Individuals store files, Discord, Email**]
- Rate(1-5) these features on how useful you would find them.
    - Searchable archive of past bridge designs
    - Database of material testing results (stick strength, glue, floss, etc.)
    - Bridge weight estimator from a submitted model
    - Structural efficiency calculation
    - Ultimate load and failure point prediction
    - Upload and visualize a team's bridge model in the browser
    - Exit documents/role handover records for executive positions
    - Onboarding guide and training material for new members
    - Private workspace for each team's own files and data
    - Competition rules and regulations reference
    - Club announcements and general information page

- Would you want your team files, results and progress to be available for future teams to learn from? **[Yes/No/Maybe]**
- If you could go back to being a new member, or if you're currently a new member, what tutorials would you want access to? **[Modeling Software, Manufacturing, Slab Building, Team Organization, Other]**

McMaster CAS Students
McMaster CAS students are needed for the Software Development Team; they will maintain and expand the toolbox. To ease the burden on them, we must elicit preferred code and documentation formats
Some questions we would like to ask are:

- When working with a system you did not develop yourself, rate(1-5) how helpful you find these documentation formats.
  - Markdown or other lightweight markup languages for README files.
  - Inline Documentation (Javadoc, Doxygen, XML Doc Comments, JSDoc, Docstrings)
  - Collaborative Wiki Formats, think online library documentation.
  - Modelling and Visual Formats (PlantUML, C4 Model)
- What naming conventions do you prefer?
  - PascalCase
  - IPascalCase (prefix interfaces with a capital I)
  - camelCase
  - _camelCase (prefix private and internal instance fields with an underscore)